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Oculomotor behaviors during visual short-term memory binding in healthy aging

Mario A. Parra[a], Grace McPherson[a], Danilo Verge[b], Nora Rotstein[c] and Gerardo Fernández[d]

[a]Department of Psychological Sciences & Health, University of Strathclyde, Glasgow, UK; [b]Chief Medical Officer and Chief Operating Officer, ViewMind Inc ., New York, NY, USA; [c]Scientific Support Team, ViewMind Inc, New York, NY, USA; [d]Chief Scientific Officer, ViewMind Inc., New York, NY, USA

ABSTRACT

The Visual Short-Term Memory Binding Task (VSTMBT) assesses the cognitive ability responsible for integrating and retaining objects' features on a temporary basis. The VSTMBT, combined with eye-tracking (ET), identified impairments in older adults with Mild Cognitive Impairment (MCI) who developed AD dementia 3 years after their baseline assessment. This study investigated whether age impacts oculomotor behaviors linked to the VSTMBT. We assessed a group of healthy young adults (18–25 years old) and a group of healthy older adults (60–83 years old) with the VSTMBT synchronized with ET. The VSTMBT required participants to detect changes across two consecutive arrays of either two or three bicolored objects. They were asked to remember the object's colors either separately (Unbound Colors Condition, UC) or combined (Bound Colors Condition, BC). We collected behavioral responses, fixation duration, saccade amplitude, and pupil dilation. Older adults remembered fewer objects but that was similar in the UC and BC conditions. Both age groups showed decreased saccade amplitudes and longer fixation duration in the BC condition, with no differential impact of age. Pupil dilation was lower in older adults, but such a behavior was equivalent across the UC and BC conditions. These null findings were confirmed by Bayesian analysis. These results suggest that binding functions and their associated oculomotor behaviors are resilient to age-related cognitive decline, highlighting the relevance of evaluating oculomotor measurements during the VSTMBT to detect the transition from normal to abnormal variants of aging earlier and more accurately.

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Introduction

Visual Short-Term Memory (VSTM) Binding refers to the cognitive ability responsible for integrating and temporarily retaining objects' features such as shapes and colors into unified representations (i.e., conjunctive bindings ,Luck & Vogel, 1997; Wheeler &

CONTACT Mario A. Parra mario.parra-rodriguez@strath.ac.uk Department of Psychological Sciences & Health, University of Strathclyde, Graham Hills Building, 40 George Street, Glasgow G1 1QE, UK
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Treisman, 2002). This function is different to that responsible for holding the association of different pieces of information (e.g., object-location, face-name, word pairs, etc.), which has been well investigated in long-term memory (LTM) as it is thought to be the backbone of episodic memory (i.e., relational binding , Mayes et al., 2007; Moses & Ryan, 2006; Tulving, 2002). The experimental literature has shown that integrating or binding information in VSTM and Working Memory (WM) requires cognitive resources (Olson & Jiang, 2002; Wheeler & Treisman, 2002). From the seminal work of Luck and Vogel (1997) and Vogel et al. (2001) later refined by Wheeler and Treisman (2002), VSTM/WM binding has been assessed using change detection tasks (Pashler, 1988). Traditionally, tasks compare the ability to hold in VSTM (see Baddeley et al., 2011 for suggestions on feature binding and the WL model) integrated features (colored shapes, bicolored objects) relative to the ability to hold individual features (colors or shapes). The former task has proved more taxing than the latter, leading to the notion that binding features in VSTM are cognitively costly.

By bringing the concept of feature binding from perception to VSTM (Treisman, 1996; Treisman et al., 1998) to explore the unit of representation of information in this cognitive system (i.e., integrated objects or features linked via additional mechanisms (Luck & Vogel, 1997; Vogel et al., 2001)), new opportunities were created to explore if the cost of forming and holding unified objects in VSTM varies across the lifespan. The literature emerging from such a question has consistently shown that this function remains preserved until old age, an observation that has held regardless of the feature types (between-dimension binding such as shape-color, Brockmole & Logie, 2013; Brockmole et al., 2008, within-dimension color–color binding, Parra et al., 2009, or even binding features of everyday objects, Hoefeijzers et al., 2017). These observations were in stark contrast to the well-known age-related associative memory decline hypothesis (Naveh-Benjamin, 2000, Naveh-Benjamin & Cowan, 2023, Old & Naveh-Benjamin, 2008), which posits that as we grow older, the ability to associate information in memory rapidly declines.

The aging literature has suggested that older adults present with a general associative memory decline regardless of the memoranda (i.e., face-name, object-location, pairs of unrelated words (Craik, 2000; Mitchell et al., 2000; Naveh-Benjamin, 2000; Naveh-Benjamin et al., 2007; Naveh-Benjamin, Guez, & Shulman, 2004). While age-related associative memory deficits have been linked to the age-related atrophy of the hippocampus (posterior Medial Temporal Lobe Network) and dysfunction of the frontotemporal network (Mayes et al., 2007; Peterson & Naveh-Benjamin, 2016), less is known about the neural underpinnings of age-insensitive VSTM Binding (VSTMB). The available evidence suggests that preservation of the anatomical integrity of regions of the anterior Medial Temporal Lobe Network such as the entorhinal and perirhinal cortex (Insausti et al., 1998) which are the correlates of context-free memory functions (e.g., VSTM binding, Bastin, 2017; Didic et al., 2011) may provide an account.

Whereas associative or relational binding declines with age, conjunctive VSTMB does not. That is, while older adults' abilities to represent in LTM and in VSTM associated information has been consistently found to be disproportionately affected relative to their ability to process the individual parts (Chalfonte & Johnson, 1996; Mitchell et al., 2000Naveh-Benjamin et al., 2007; Naveh-Benjamin, Guez, Kilb, et al., 2004; Naveh-Benjamin et al., 2003), such a dissociation has been rarely found for conjunctive information held in VSTM (i.e., the decline in processing shape-color conjunctions has not been larger than that seen in processing each feature separately, see for example, Brockmole & Logie, 2013). Such age-preserved VSTMB abilities have been extensively replicated (Brockmole et al., 2008; Hoefeijzers et al., 2017; Parra et al., 2009; Rhodes, Parra, et al., 2015; van Geldorp et al., 2014; Yassuda et al., 2019).

These observations are important because a memory function that remains preserved across the lifespan can help identify older adults who are at risk of dementia earlier and more accurately (Logie et al., 2015; Parra, 2022). For example, Didic et al. (2011), proposed that regions of the anterior Medial Temporal Lobe network, which support context-free memory functions (Bastin et al., 2019; Mayes et al., 2007; Wolk et al., 2008) and are anatomically preserved across the lifespan (Insausti et al., 1998) are affected by pathological aging variants such as Alzheimer's disease years to decades prior to symptoms onset (see Juottonen et al., 1998 for neuroanatomical evidence). Such evidence can help explain why the VSTMB Task (VSTMBT) has proved sensitive to the preclinical stages of Alzheimer's disease (AD) (Forno et al., 2022; Koppara et al., 2015; Parra, et al., 2010a; Parra et al., 2022). The VSTMBT assessing color–color binding, combined with eye-tracking (ET), has identified impairments in older adults with Mild Cognitive Impairment (MCI) who have developed AD dementia 3 years after their baseline assessment Parra et al., 2022a). Using a version of the VSTMBT reported by Parra et al. (2009) found that, behaviorally, age affects VSTM in general, but there was no evidence of specific age-related VSTMB deficits. In this study, we investigated whether the age-insensitive property of the behavioral outcome measure holds for physiological measures drawn from eye-tracking.

ET has become a reliable neuroscience method to unveil neural correlates of complex cognitive functions and behaviors in healthy individuals and those affected by neuropsychiatric diseases (Duchowski, 2017). The technique has been traditionally used to explore the integrity of the motor system via directed eye movements (e.g., pro-saccades, anti-saccades), and more recently oculomotor behaviors associated with cognitive and neuropsychological task performance (Duchowski, 2017; Dunn et al., 2023; Hodgson et al., 2019). The literature confirms that age impacts all these domains. Older adults typically exhibit longer saccade latencies (Yang & Kapoula, 2006), slower saccade speed (Abel et al., 1983), less accurate eye movements due to limited precision of saccades which often require corrective saccades (Irving et al., 2006), longer fixation durations, indicating a need for more time to process visual information (McCarley et al., 2012), reduced ability to smoothly follow a moving object leading to more frequent corrective saccades during pursuit (Moschner & Baloh, 1994), their pupillary responses to cognitive and visual stimuli may be slower and less pronounced (Winn et al., 1994), and can show instability or drift when maintaining a steady gaze, which further affects their ability to fixate steadily on a target (Leigh & Zee, 2015). In the field of pathological aging, particularly that pertinent to neurodegenerative diseases, the study of ET metrics, together with neuropsychological tasks, is shedding new light on the mechanisms underlying disease expression. For example, using the CANTAB Spatial Working Memory task (Hodgson et al., 2019), found that patients with Parkinson's disease make significantly fewer forward planning fixations consistent with a deficit in strategic planning. The authors argued that refixations might reflect weak memory traces or partial forgetting of information and suggested that eye movement metrics (e.g., refixations) might provide a more sensitive measure of impairments than traditional behavioral scores. A potential caveat to this methodology is that the neuropsychological task combined with ET should ideally be insensitive to normal aging, which is not the case for the CANTAB Spatial Working Memory task (Brown, 2016) and several other tasks used for the early detection of age-related diseases such as those that cause dementia (e.g., CANTAB Pair Associates Learning Task, de Jager et al., 2002). The traditional approach to address the influence of such a confounding variable has been to develop age-adjusted normative datasets. However, recent studies suggest that such norms may not always provide a reliable context to ascertain normality or lack thereof. This could be because the heterogeneity that characterizes the older population might not always be fully captured (Mungas et al., 2009, 2010) or simply because the very tests used to select participants to develop the norms are insensitive to early still subclinical stages of pathological aging such as AD (Bos et al., 2018; Parra et al., 2024)

Taken together, the above-reviewed studies and the predictions on the role that ET methods will play as biomarkers for neurodegenerative disease (Lim et al., 2016; Marandi & Gazerani, 2019) they endorse the need to subject the new methodology (i.e., VSTMBT + ET) to an aging study. By demonstrating that this new neurocognitive biomarker is unaffected by the normal course of aging, we can grant further reliability to the evidence drawn from studies targeting age-related diseases. That was the aim of the present study. We focused on three ET metrics which we have proved sensitive to detect the presence of AD dementia and reliably predict its risk. We analyzed fixation duration, saccade amplitude, and pupil dilation in tasks involving individual and integrated features using different memory loads (Fernández & Parra, 2021, Parra et al., 2022b; Fernandez et al., 2022; Fernández et al., 2019). Based on the accumulated evidence from behavioral studies, we predicted that a general age-related decline could be observed on both behavioral and ET metrics, but this would not be specific to feature binding.

Materials and methods

Participants

Data was obtained from 50 participants, from two different age categories, a younger adult group that involved 25 individuals between 18 and 25 years old, with a mean age of 23 (SD = 1.70) and the older adult group that comprised 25 individuals between 60 and 83 years old, with a mean age of 72 (SD = 6.10). The younger group had spent significantly more years in education compared with the older sample, with a mean of 17 years (SD = 1.00) and 14 years of education (SD = 3.30), $t(48) = 5.23$, $p < .001$, respectively. However, it is worth noting that the VSTMBT has proved insensitive to the level of education of those assessed (Parra et al.,, 2011; Yassuda et al., 2019). A priori power analysis was conducted for a between-subjects design with a significance level of 0.05, assuming an effect size (difference in means) of 0.5 and a standard deviation of 1. The analysis indicated that with a sample size of 25 per group, the test would have a power of 0.86 to detect a statistically significant difference between the groups. A pre-screening demographic questionnaire and perceptual tests (see Parra et al., 2010b) ruled out color vision or perception problems, with all participants having normal or corrected to normal visual acuity. These screenings were undertaken to rule out the possibility that poor performance on the VSTMBT could have resulted from visual or perceptual difficulties (Parra et al., 2011). Participants were recruited through word of mouth and from the University of Strathclyde

Panel of Volunteers. This study was approved by the School of Psychological Sciences and Health Ethics Committee of the University of Strathclyde (Ref: 37/07/11/18/A).

Neuropsychological assessments

Older adults completed the Mini Mental State Examination (MMSE, Folstein et al., 1975). All the participants scored 26 or above, suggesting healthy cognitive functioning (M = 28.76, SD = .97) (Law et al., 2012). Additionally, all participants completed the Test of Premorbid Functioning (TOPF, UK, Wechsler, 2011) to obtain a measure of their verbal IQ (Younger adults: $M = 112.14$, SD = 24.08; Older adults: $M = 116.38$, SD = 9.56, $p = 0.416$).

The VSTMBT and eye movements assessment

The task design was similar to that used previously (Fernandez et al., 2018). Participants were presented with arrays of objects, 60 cm away from a 21″ LCD monitor screen using a 3 × 3 virtual grid. Participant movements were restricted by using a chin rest. Eye movements were recorded using Gazepoint with a sampling rate of 150 hz. All recordings and calibrations were binocular, and the task began after successful calibration. The change detection paradigm was previously described by Parra et al. (2009) and used by Fernandez et al. (2018). Each object was defined by a shape and a frame area, making up 50% of the object's surface each; both were filled with a color (see Figure 1). Both groups completed the same trials during the task. Trials began with a fixation screen (a cross) shown for 250 ms, followed by a study display for 2000 ms, an unfilled retention interval of 900 ms and finally the test display, presented for 4000 ms (Figure 1). In half of the trials, objects on both displays were the same, whereas, in the other half, objects displayed different colors in the test display. The location of objects was randomly changed within a 3 × 3 virtual grid to ensure items' location was an uninformative feature. Participants were requested to detect whether the study and test displays consisted of the "same" or "different" objects by selecting either "S" or "D" on the keyboard in front of them.

Four experimental conditions were used (Figure 1). The Unbound Colours condition (UC) involved two or three bicolored objects with the shape and frame areas showing different colors. In the "different" trials, the colors from either the shape or the frame area (with 50% probability) of two objects of the test display were replaced by new colors from the available set that had not appeared in the study display. In the Bound Colors condition (BC), in the "different" trials, two objects swapped one color, either from the shape or the frame area (with 50% probability). In the UC trials, participants were told to focus on the colors and not their combinations, as new colors would be introduced. In the BC trials, participants were told to focus on the colors and their combinations within objects, as no new colors would be introduced on the second screen, but the same studied colors could be recombined in different objects. Both conditions involved two set sizes, two or three stimuli assessed memory load and the impact this could have on both oculomotor behavior and VSTMB performance. Task conditions (BC and UC) and Set Sizes (two or three) were blocked. Each block comprised 10 practice trials, followed by 32 test trials (i.e., 32 × 4 = 128, 16 same and 16 different for each block). Trials were fully randomized, and conditions were blocked and

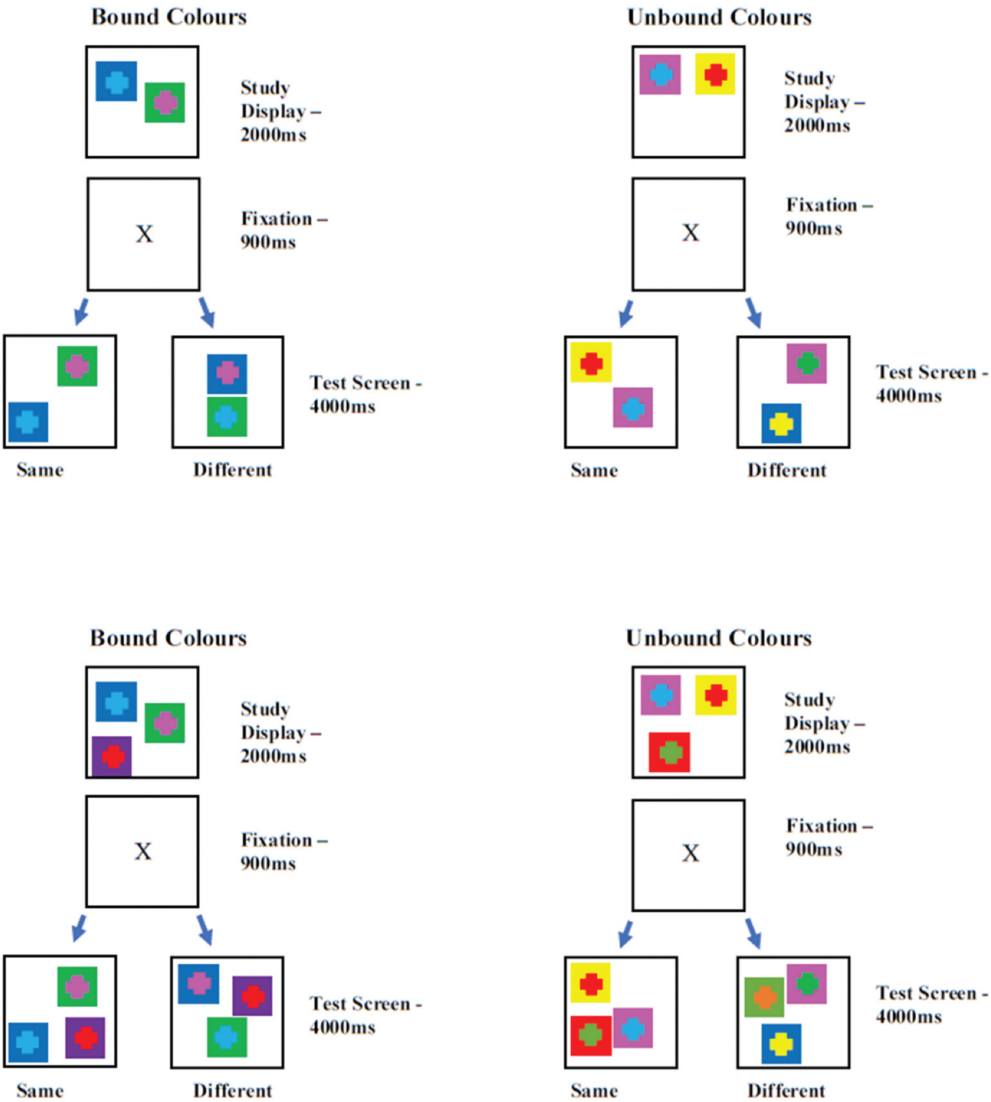

Figure 1. An example trial of the VSTMBT for the Bound Colour (BC) and Unbound Colour (UC) condition for set sizes two (upper panel) and three (lower panel).

counterbalanced across participants. During the task, accuracy (i.e., percentage of correct recognition) was recorded for each condition. In addition, three oculomotor measures were considered: fixation duration, saccade amplitude, and pupil dilation (Fernandez et al., 2018). Operationally, increased fixation durations denote more time needed to process the fixated information, indicating fewer cognitive resources available, increased task demands, or a combination of these. Smaller saccade amplitudes are associated with higher cognitive effort, indicating that cognitive resources are redirected toward processing rather than exploring visual stimuli. Increased pupil diameter is known to indicate increased cognitive effort/ load.

Procedures

An information sheet was provided, and all participants signed a consent form. Following full consent, a demographic questionnaire was completed including questions relating to current medication, diagnosis of neurological/non-neurological conditions, years of education and vision/auditory abilities. The Perceptual Binding test was then administered (Parra et al., 2010b) which comprised 10 trials. Participants need to achieve 80% accuracy to be eligible for the memory test. None was excluded in the present study. Individuals were then asked to place their chin on a chin rest, while their eyes were calibrated with a standard 13-point grid for both eyes. Subsequently, participants were presented with printed flash cards showing examples of the test arrays, one for each of the four conditions. These examples were presented individually before each condition to avoid confusion. This allowed individuals to familiarize themselves with the test instructions during the practice trials. Participants then began the experiment. They completed all four conditions, with verbal instructions given at the start of each new condition. Following the VSTMBT, both age groups completed the TOPF (Wechsler, 2011), whereby their responses were recorded for scoring purposes and IQ was calculated. In addition to the TOPF, the older group completed the MMSE (Folstein et al., 1975), in order to screen for MCI. Participation in the study took between 45 and 60 min.

Statistical analysis

We subjected the data to a three-stage analysis. First, we conducted the classical General Linear Model for a mixed design with between and within-subjects factors. This has been the model traditionally used to explore age-related decline using the VSTMB (Brockmole et al., 2008; Parra et al., 2009). To that aim, we defined a 2 (Age Group) × 2 (Condition) × 2 (Set Size) mixed ANOVA model. Behavioral and oculomotor data were entered into this model separately. Main effects, two-way and three-way interactions, are reported (see Supplementary Material). Second, a Bayesian analysis (BF_{01} H0/H1) was carried out as our primary hypothesis sought evidence supporting the null hypothesis (i.e., no differential age-related decline). To that aim, we calculated the Binding Cost by adapting the formula proposed by Forno et al. (2022) and Parra et al. (2024) (i.e., Binding Cost = ((UC − BC)/UC) * 100). This measure informs how taxing the BC condition is relative to the baseline condition (UC). We calculated such a cost for all the study variables. Finally, we performed a whole-sample regression analysis to explore whether the mean accuracy on the VSTMBT could be predicted by mean oculomotor behaviors collapsing across experimental factors (Condition and Set Size). To that aim, Bayesian regression models were used for each predictor (see Supplementary Material S2 for a traditional stepwise regression model). It is not uncommon to encounter works that have combined or recommend the combination of Bayesian-Frequentist models to overcome the limitations and complement the strengths of each statistical approach (Arning et al., 2024; Fryer et al., 2023). Statistical analyses were performed in SPSS version 28.0.1.1 (IBM-Corp, 2021).

Results

Behavioral data

Mean behavioral data during the VSTMBT across conditions is shown in Table 1. The performance of the older group was similar to that of the younger group with Set Size 2 on both the BC (92% and 95% of correct responses for older and younger participants, respectively) and UC condition (92% and 96%, respectively). In contrast, with Set Size 3, performance was lower for the older than for the younger group, both on the BC (74% and 81%, respectively) and the UC condition (78% and 86%, respectively).

The Mixed ANOVA revealed a significant main effect of Group, with higher accuracy in the younger relative to the older group. The effect of Set Size was also significant, with better accuracy for Set Size 2 than Set Size 3. In addition, a significant main effect of Condition was found, with BC yielding lower accuracy than UC. The key Group × Condition interaction was non-significant (see Supplementary Material for detailed outcomes of Mixed Models for Behavioral and Oculomotor Variables). As the Set Size did not modify the lack of Group × Condition interaction across outcome measures (i.e., Behavioral and Oculomotor Variables), data were collapsed across this factor, and such averages were used to calculate the Binding Cost, which entered Bayesian analyses. Operationally, the Binding Cost indexes the cognitive effort when processing bound colors (i.e., BC) relative to individual colors (i.e., UC). Therefore, from a behavioral perspective, an increased Binding Cost suggests that the participant experienced the BC condition as a more challenging task. For the oculomotor behaviors, this operationalization varies. For example, an increased Binding Cost on Fixation Durations would suggest that more time was needed to process stimuli during the BC than the UC condition, indicating a more challenging nature of the former condition. An increased Binding Cost on Saccade Amplitude would inform on smaller amplitudes during the UC than the BC condition, indicating that more cognitive resources were deployed toward the former condition. Finally, an increased Binding Cost on Pupil Dilation would result from larger pupil diameters in the BC than in the UC condition, indicating increased cognitive efforts during the former.

As observed during the Mixed ANOVA model, processing BC was more costly than processing UC, an effect that was similar across age groups. The Bayes Factor confirmed that we are 4.04 times more likely to find an age-related behavioral effect of the Binding Cost like the one reported in Table 2 under the null hypothesis than under the alternative hypothesis (i.e., moderate evidence for H0), with a 95% Credible Interval further supporting this notion (i.e., –6.04/3.32). However, the direction of this effect was driven by a larger cost in the younger than in the older group.

Oculomotor measurements

Fixation duration

When Fixation Duration entered the Mixed Model, the lack of two-way interaction (Group × Condition) and three-way interaction (Group × Condition × Set Size) reported for behavioral data were also observed (see Supplementary Material). Fixation Duration increased from Set Size 2 to 3 (significant main effect of Set Size) and was longer in

Table 1. Mean, standard deviations of demographic, oculomotor, and behavioral measurements across Group, Set Size, and Condition.

	Demographics						Oculomotor Measurements			Behavioral Measurement
Age Group	Age	Education (Years)	TOPF IQ[1]	MMSE[2]	Set Size	Condition	Saccade Amplitude	Pupil Dilation	Fixation Duration	Accuracy (%)
Younger ($n = 25$)	23 (1.70)	17 (1.00)	112.14 (24.08)	–	2	BC UC	2.44 (0.52)	3.58 (0.51)	7.07 (.16)	95.02 (5.90)
							2.67 (.41)	3.49 (.48)	7.10 (.17)	96.01 (3.88)
Older ($n = 25$)	72 (6.10)	14 (3.30)**	116.38 (9.56)	28.76 (.97)	2	BC UC	2.69 (1.11)	2.89 (.67)	7.10 (.23)	92.14 (5.50)
							2.99 (1.12)	2.88 (.67)	7.13 (.20)	92.01 (8.38)
Younger					3	BC	2.55 (0.37)	3.67 (.51)	7.27 (.17)	80.75 (7.95)
						UC	2.63 (.48)	3.61 (.49)	7.21 (.16)	86.13 (6.87)
Older					3	BC	3.01 (1.27)	2.84 (0.75)	7.22 (.45)	73.50 (11.01)
						UC	3.13 (.93)	2.90 (.69)	7.15 (.32)	78.00 (9.31)

[1]Test of Premorbid Functioning, UK. [2] Mini-Mental State Examination.
*$p < .05$, **$p < .0$, ***$p < .001$.

Table 2. Bayesian independent-group analysis using the binding cost for behavioral and oculomotor variables.

Binding Cost Variables	Mean (SD) [a]	Mean Difference	BF [b]	t (df = 48)	p-value	95% Credible Interval
Behavioral Accuracy	Young: 3.37 (6.51) Older: 2.01 (9.34)	−1.36	4.04	−0.60	0.553	(−6.0/3.3)
Fixation Duration	Young: −0.48 (3.09) Older: −0.27 (3.54)	0.20	4.64	0.22	0.829	(−1.7/2.1)
Saccade Amplitude	Young: 5.41 (9.86) Older: 7.18 (14.09)	1.77	4.21	0.52	0.608	(−5.3/8.8)
Pupil Dilation	Young: −1.94 (1.59) Older: 0.63 (8.47)	2.57	1.77	1.49	0.142	(−0.9/6.1)

[a]Means of the Binding Cost (see text for the interpretation of these measures); [b] BF: Bayes factor, null versus alternative hypothesis (BF01).

the BC than in UC condition in Set Size 3 but not in Set Size 2 (a significant Condition × Set Size Interaction). However, all these effects were independent of age.

The Bayes Factor confirmed that we are 4.64 times more likely to find an age-related effect of the Binding Cost on Fixation Duration as large as the one observed under the null (i.e., moderate evidence for H0) than the alternative hypothesis (95% Credible Interval = −1.73/2.14).

Saccade amplitude

When the Saccade Amplitude entered the Mixed Model, the lack of two-way interaction (Group × Condition) and three-way interaction (Group × Condition × Set Size) were also observed (see Supplementary Material). The BC condition led to lower saccadic amplitude than the UC condition, but again, such a discrepancy did not differ across groups. The Bayes Factor confirmed that we are 4.21 times more likely to find an age-related effect of the Binding Cost on the Saccade Amplitude as large as the one observed under the null (i.e., moderate evidence for H0) than under the alternative hypothesis (95% Credible Interval = −5.30/8.85).

Pupil dilation

As for Behavioral Accuracy, Pupil Dilation also proved sensitive to normal aging. Such sensitivity was independent of the other investigated factors and did not lead to any significant interaction between them. The Bayes Factor confirmed that we are 1.77 times more likely to find an age-related effect of the Binding Cost on Pupil Dilation as large as the one observed under the null (i.e., anecdotal evidence for H0) than the alternative hypothesis (95% Credible Interval = −0.98/6.13). These results suggest that although Pupil Dilation as an index of cognitive load declines with age, its variations are not specific to binding functions in VSTM.

Predicting VSTMB accuracy via oculomotor behaviors

Bayesian regression analyses were conducted with mean Behavioral Accuracy as the dependent variable and mean Fixation Duration, Saccade Amplitude, and Pupil Dilation as predictors (Table 3). The model, including Fixation Duration, showed 0.173 times more

Table 3. Summary of the Bayesian regression analysis for oculomotor behaviors predicting VSTM accuracy.

Variable	BF [a]	R	R^2	*F*	*p*-value	95% Credible Interval
Fixation Duration	0.137	0.096	0.009	0.443	0.509	−11.48/5.77
Saccade Amplitude	1.104	0.305	0.093	4.91	0.031	−4.68/-0.23
Pupil Dilation	8.801	0.411	0.169	9.78	0.003	1.31/6.04

[a]BF: Bayes Factor, alternative versus null hypothesis (BF10).

support in the data for H1 (i.e., no evidence for H1), which led to a non-significant effect and a 95% Credible Interval including 0. The model, including the Saccade Amplitude, showed 1.104 times more support in the data for H1 (i.e., anecdotal evidence for H1), which led to significant results. Finally, the model, including Pupil Dilation, showed 8.801 times more support in the data for H1 (i.e., moderate evidence for H1), which led to significant results. The traditional stepwise regression model's results aligned with these outcomes (see Supplementary Material S2).

Discussion

The present study set out to investigate the hypothesis that a general age-related decline would be observed on both behavioral and ET metrics drawn from the VSTMBT, but this would not be specific to feature binding. This hypothesis was tested for behavioral and ET metrics using Bayesian and traditional (see Supplementary Material) analysis. For the former, we implemented, for the first time, the calculation of the Binding Cost for Behavioral and Oculomotor variables. Comparing the Binding Cost across groups is an additional and more informative way of investigating potential interactions. Our results show that healthy older and younger adults exhibited high performance across all tasks at both set sizes. In addition, both age groups had similar Saccade Amplitudes and Fixation Durations, whereas younger adults had larger Pupil Dilation, independently of Set Size and Condition. Both age groups showed a lower accuracy for BC than for UC, as denoted by a positive Binding Cost, which corresponded with shorter Saccade Amplitude and longer Fixation Duration. We discuss these findings in turn.

The VSTMBT used in this study evaluates the ability to integrate color information within unified representations (BC) and compares this with the ability to process single colors (baseline UC) (Parra et al., 2009). This has consistently demonstrated that processing BC is more resource-demanding than processing UC, a cost that is independent of age or brain health status (i.e., patients with or at risk of Alzheimer's disease dementia (ADD) show a much larger cost, Parra et al., 2011). Older adults showed high accuracy on UC and BC, similar to that of younger adults, further supporting the notion that, behaviorally, binding within-dimension features in VSTM was found to be preserved in healthy aging (in line with Parra et al., 2009). Both age groups performed similarly on the UC and BC conditions at a set size 2. A slight, although significant, decrease in performance was shown for older adults at set size 3, implying younger adults outperformed older adults when memory load increased, as previously reported (Yassuda et al., 2019). These differences might arise from the marked decrease in VSTM capacity with age; VSTM has been shown to peak in the early twenties and then decline, with a sharp decrease in capacity (Brockmole & Logie, 2013). Previous studies using change detection or reconstruction tasks showed a lack of effect of aging on the cost of feature binding (Brockmole et al., 2008; Killin et al., 2017; Parra et al., 2009), even for features of everyday objects (Hoefeijzers et al., 2017). Of note, older participants in this study remembered fewer objects, but that was true regardless of the task condition (i.e., UC or BC). Although memory load amplifies the effects of age due to reduced storage and processing capabilities within VSTM and Working Memory, the ability to process the binding between colors within unified representations was found to be preserved in healthy older adults. In fact, a larger Binding Cost was observed for the younger than the older group, leading to moderate evidence for H0 in our Bayesian analysis.

Evaluation of Fixation Duration and Saccade Amplitude has proved effective not only in discriminating between controls and patients with MCI but also in accurately predicting those who will convert to ADD (Parra et al., 2022b). We showed here that saccades were smaller in the BC than in the UC condition, an effect that was more pronounced in the younger group (as indicated by the Binding Cost). This discrepancy yielded moderate evidence for H0, suggesting that, relative to older adults, younger adults, on average, visually scanned BC more than UC. Together with the behavioral outcomes, these results suggest that such an amount of visual scanning was sufficient for older adults to process the binding between colors, hence resulting in a smaller behavioral cost relative to that seen in the younger group.

Saccadic movements are influenced by a range of cognitive processes, notably those concerning attention and working memory, and provide a tool to analyze the functionality of the saccade-generating circuit in the brainstem (Sparks, 2002). A previous study by Parra et al. (2022b) showed that MCI patients at risk of ADD present with an increased Saccade Amplitude relative to controls, thus suggesting that as older adults depart from the typical aging trajectory, meeting the taxing nature of the VSTMBT becomes more challenging. The VSTMBT engages regions within the ventral visual stream, which is responsible for object identification and recognition, while the dorsal stream relates to object spatial location and is instrumental in saccade generation (Parra et al., 2014; Irwin et al., 2004). Both visual streams appear to be preserved in healthy aging but are damaged in AD (Irwin et al., 2004; Deng et al., 2016). Our results are consistent with the proposal that saccadic eye movements at the service of VSTMB performance are resilient to the effects of normal aging. The information encoded in VSTM informs target selection before saccade initiation (Van der Stigchel and Hollingworth, 2018). Using the VTSMBT, we have previously demonstrated that encoding is the memory phase most affected by ADD (Fernández & Parra, 2021). The fact that such memory processes appear to be spared by healthy aging grants this tool reliability as a marker to aid the detection of ADD. This represents a significant step forward in the neurocognitive assessment of dementia, as most available assessment tools need to control for the effect of this demographic factor (Parra, 2022), a statistical procedure that has proven prone to interpretation errors (Mungas et al., 2009, 2010).

Fixation Duration is sensitive to both memory load and processing load, as it increases with the number of targets and task difficulty (Meghanatan et al., 2015. The scarce evidence available suggests that older subjects display longer fixation durations in a visual search task (Porter et al., 2010). We observed that Fixation Duration was similar in older and younger adults, with both age groups showing longer fixation durations with increased set size. The Cost of Binding on Fixation Duration showed that, on average (across set sizes), younger adults fixated more on BC than on UC, relative to older adults, leading to moderate evidence for H0. Although this discrepancy across groups was small (as suggested by frequentist analysis), they suggest that, when compared to the younger group, the older group did not find the BC more challenging than the UC. Hence, whereas we confirmed the sensitivity of Fixation Duration to memory load, we failed to prove its sensitivity to the effects of normal aging. That does not seem to be the case for abnormal aging variants such as ADD. Such patients consistently present with shorter fixation durations in the BC condition when compared to healthy older adults (Fernandez et al., 2018). Therefore, as for Saccade Amplitude, the evaluation of Fixation Duration provides an additional oculomotor biomarker to reliably discriminate between ADD and healthy aging. We have observed that as cognitive demand increases, fixation durations tend to increase, and saccades become smaller, thus suggesting that more cognitive processing is required per fixation before moving on to the next visual target.

Intriguingly, Pupil Dilation was the only oculomotor behavior that showed age effects. Younger adults had larger pupil diameters than older adults, but that was true in both task conditions. In fact, Pupil Dilation was the oculomotor behavior that provided the strongest evidence for H0 (i.e., for Binding Cost), and the only covariate retained by a regression model aimed at predicting Behavioral Accuracy (see Supplementary Material S2). Pupil size correlates with task difficulty, attentional focus and mental effort (Hess and Polt, 1964; Eckstein et al., 2017) and has been linked to better performance (Goldinger et al., 2012; Alnaes et al., 2014). Younger adults in our study seem to follow this pattern. Although the differences are non-significant, they showed subtly larger values for BC than for UC, and these values increased slightly with increasing set size, changes consistent with higher cognitive demand. In contrast, the older group had very similar pupil sizes in both conditions, implying the ability to adjust to higher cognitive demands might fluctuate with age. Nevertheless, this was not specific to feature binding in VSTM. Pupillary responses reflect the processing load and are underpinned by subjects using strategies that reduce this load (Meghanatan et al., 2014). This may explain the observation of similar Pupil Dilation across task Conditions and Set Sizes, particularly in the older group. It is worth noting that participants in this study, particularly those from the older group, were recruited from a volunteer panel. Parra et al. (2022a) highlighted the potential influences of such selective samples on the data (see also Ganguli et al., 1998). Older adults involved in such panels are regularly engaged in research and frequently undergo cognitive testing, which grants them familiarity with testing procedures and additional cognitive reserves and resilience (Infurna et al., 2016), thus representing a somewhat biased subsample of the relevant population.

An alternative explanation linked to theories from cognitive psychology (i.e., cost-free vs costly feature binding, Luck & Vogel, 1997; Vogel et al., 2001; Wheeler & Treisman, 2002) is that although behaviorally binding colors in VSTM have a cost, such a cost may not reflect significantly different neurobiological demands. Using fMRI, Parra et al. (2014), observed in healthy young adults who performed a shape-color VSTMBT that increasing memory load (i.e., from set size 2 to 4) did not lead to a significant discrepancy in neural recruitment and both set sizes activated the same brain regions known to be relevant for VSTMB. As long as the memory load remains within capacity (i.e., 4 items, Cowan, 2001), the neural resources needed to reach a high level of performance might not vary dramatically, pointing toward a brain economy process which might reduce the cost of feature integration in VSTMB (Luck & Vogel, 1997, 2013; Oberauer & Eichenberger, 2013; Vogel et al., 2001) and tackles the binding problem of the human brain (Treisman, 1996). This, together with the observation of the sensitivity of this function to ADD, suggests that Pupil Dilation may be a suitable marker for identifying subtle fluctuations in cognitive processing (i.e., even at low levels of memory load) as one departs from the normal aging trajectory (Fernández & Parra, 2021). A further advantage of the VSTMBT is that it has proved insensitive to the cultural background of those assessed, as the level of education does not affect performance during binding tasks (Calia & Parra, 2024; Martínez-Flórez et al., 2024, Parra et al., 2011; Yassuda et al., 2019). This further endorses the psychometric robustness of this task as a detection tool for ADD cross-culturally (Della Sala et al., 2016).

Why are the ET metrics here investigated following the same behavioral pattern previously reported for the VSTMB, despite a large and consistent literature suggesting that all such metrics are age-sensitive? To answer this question, we referred to the work recently published by Hodgson et al. (2019). The authors proposed an updated model for oculomotor behaviors during neuropsychological tasks. Their second update, which is relevant to our findings, is what the authors called a "trigger function." They suggest that when task-critical information faces challenges entering memory (i.e., encoding) or is degraded, the oculomotor system is triggered to fixate the to-be-remembered objects further. That is, VSTM and WM can draw support from the eye movement system to address additional needs. The fact that behavioral studies have consistently shown that VSTMB abilities remain preserved in older age can suggest that by calling up such support mechanisms, older adults may compensate for the age-related decline, bringing performance to a level comparable to that of younger adults (see for example Grady, 1998; Grady & Craik, 2000). The literature supporting the presence of compensatory mechanisms mitigating the impact of cognitive aging has grown significantly over the last decades (Cabeza et al., 2018; Reuter-Lorenz & Park, 2014; Stern, 2021). We have confirmed that the sensory input stage (perceptual binding) is intact in these participants and have shown that their memory and oculomotor behavior are also preserved. It follows that normal aging spares the complex sensorimotor transformations underpinning these neurocognitive systems and the resources needed to support them. Hodgson et al. (2019) acknowledged that a shortcoming of eye movement testing in the context of neuropsychological assessments and diagnosis of neurological and psychiatric diseases is the low specificity for particular disorders. The results from the present study provide further evidence which supports why this new method that combines age-insensitive cognitive and ET measures has consistently achieved high levels of sensitivity and specificity for AAD (Fernandez et al., 2022; Fernández et al., 2019; Fernández et al., 2019, 2021; Parra & Fernandez, 2023; Parra et al., 2022b). Nevertheless, it is worth noting that our current findings need to be interpreted in the context of ET measures drawn from a VSTMBT that relies on the whole-display Change Detection Paradigm (Rhodes, et al., 2015; Wheeler & Treisman, 2002). Age-related working memory binding deficits have been reported using other tasks which may tap into different neurocognitive processes (e.g., recall rather than recognition, familiar instead of unfamiliar items, and different attention loads, Brown & Brockmole, 2010; Manga et al., 2021). Future work will be needed to demonstrate whether this age-insensitive memory ability is taskinvariant. An additional limitation is the nature of the sample collected for this study, which was recruited from a volunteer panel. Future studies should explore the oculomotor behaviors during VSTMB, which are investigated here in more diverse samples drawn from the general population.

In conclusion, the present study goes beyond previous investigations by coupling the analysis of oculomotor measurements to the VSMBT, thus providing further insight into age-related changes in a healthy population. Our results suggest that a decline in oculomotor behaviors such as fixation duration, saccade amplitude, and pupil dilation during VSTMB are not generalizable characteristics of age-related memory decline. Since these behaviors are impaired early during ADD progression, their analysis might provide a sensitive tool for early detection of this disease.

Abbreviations

AD: Alzheimer´s disease;
BC: bound condition;
MCI: mild cognitive impairment;
MMSE: Mini Mental State Examination;
SCD: subjective cognitive decline;
TOPF, Test of Premorbid Functioning;
UC: unbound condition;
VWM: visual working memory;
VSTM: visual short term memory;
VSTMB: visual short term memory binding;
WM: working memory.

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Disclosure statement

GF is currently the CSO of ViewMind. MAP is currently a consultant for ViewMind as a Neuroscientific Officer. NV is the Chief Medical Officer and Chief Operating Officer.

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Author contributions

MAP and GF conceived the study and developed the protocol. GM collected the data and ran the analysis under the supervision of MAP and GF. All the authors contributed to drafting the manuscript.

Data availability statement

The data used to generate the outcomes reported here could be available partially or complete on requests made to the corresponding author.

Ethical statement and consent

The authors assert that all procedures contributing to this work comply with the ethical standards of the relevant national and institutional committees on human experimentation and to the principles of the Declaration of Helsinki. The Ethics Committee from the School of Psychological Sciences and Health of the University of Strathclyde reviewed and approved this study [37/07/11/18/A]. All participants provided written informed consent following a detailed description of the study.

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